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Final report on Cubesat propulsion unit

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Abstract

Over the last two decades, CubeSats have dramatically lowered access costs to space, allowing a greater participation of the scientific community in space science. Even though the market is constantly growing, CubeSats do not possess a dedicated propulsion system to change their orbit at will after release, and they are mostly confined to their release orbit. This study aims to address this limitation by presenting a dedicated bi-propellant liquid propulsion system for 1U nanosatellites, allowing transfers with a single or multiple maneuver up to a total Δv of 600 m/s.

Chapter 1 will start by detailing the thrust unit: from propellant choice to combustion chamber and nozzle sizing, including design considerations and relevant calculations. Following, chapter 2 will describe the design for the injector system, feed lines and storage tanks, and it will also contain a mass and volume evaluation for the final design, including calculation of the center of mass (CM) position. Concluding, a final evaluation of performance losses, followed by final considerations on the design and possible failure modes, makes up chapter 3.

Acronyms

N_2H_4 Hydrazine

CC Combustion chamber

CEA NASA's Chemical Equilibrium with Applications code

CG Center of gravity

CM Center of mass

LEO Low Earth Orbit

MMH Monomethyl Hydrazine

NTO Nitrogen tetroxide

UDMH Unsymmetrical Dimethyl Hydrazine

Non-dimensional quantities

ϵ Expansion area ratio

k Specific heat ratio

o/f Oxidizer-to-Fuel ratio

c_f Thrust coefficient

$c_{f_{vac}}$ Thrust coefficient in vacuum

C_d Discharge coefficient

Re Reynolds number

Re_{pipe} Reynolds number of the propellant along the pipes

λ Distributed losses coefficient

Physical quantities

ΔP_{inj} Injector pressure drop [Pa]

ΔV Velocity change [m/s]

$\dot{m}$ Mass flow rate [kg/s]

μ_{amb} Viscosity at ambient temperature [Pa*s]

ρ Density [kg/m^3]

A_t Throat area [m^2]

C^* Characteristic velocity [m/s]

I_{sp} Specific impulse [s]

L^* Characteristic length [m]

M Molar mass [kg/kmol]

P_c Combustion chamber pressure [bar]

S Chamber structural stress [Pa]

T_c Combustion chamber temperature [K]

v_{pipe} Velocity along the pipes [m/s]

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Chapter 1

Thrust chamber

The propulsion system is required to provide a Δv of 600 m/s to a payload that has a mass of 1.33kg, contained in a single Cubesat Unit (or "1U", total volume 1dm^3), that cannot be subjected to an acceleration greater than 3g. Furthermore, the system must be suitable for long term missions, and it is a strict requirement that not only the final system complies with CubeSat standards, but that every single component as well be restricted to 1U at most, both in terms of mass and volume. All design requirements and constraints are summarized in Table 1.1. As an additional parameter of performance, it was decided to set the maximum value for the system's dry mass as two times the payload's.

Requirements	**Storage constraints**	**Mission constraints**
Payload: 1U, 1.33kg	Volume: 1U (1dm^3)	Mission duration: months
Δv=600m/s	Mass: 1U (1 33kg)	Maximum payload acceleration: 3g

Table 1.1: Summary of design constraints

For this system, volume constraints are the most restrictive since each component must fit into a 1l-cube: therefore, the team started by selecting the nozzle's exit diameter. To allow for the nozzle's wall thickness and structural parts near the nozzle's end, the diameter was chosen as 90mm. Being in vacuum, an area ratio of at least 40 is desired [2], but smaller throat radii suffer more significantly from boundary layer losses. Aiming for high expansion efficiency and small losses at the throat, the team selected $\epsilon = 100$, obtaining a throat diameter of 9mm.

For long term missions it is useful to have propellants that are liquid at the equilibrium temperature of the spacecraft, to avoid the need for thermal control. In LEO, which is the primary type of orbit used for CubeSats, the equilibrium temperature is estimated to be around 297K [1]. To avoid including an ignition system which would increase complexity, the team evaluated hypergolic couples for propellant trade off. The selected oxidizer was Nitrogen Tetroxide (NTO) and the three fuels studied were Hydrazine (N_2H_4), Monomethyl Hydrazine (MMH) and Unsymmetrical Dimethyl Hydrazine (UDMH). Properties for all four propellants [4, 5] are shown in Table 1.3.

d_e (mm)	90
d_t (mm)	9
$a_{max}(m/s^2)$	29.43

Table 1.2: Fixed design parameters

The three propellant couples generate exhaust gases with a k between 1.20 and 1.30 [3]: since the thrust coefficient c_f and the ratio $\frac{P_c}{P_e}$ are functions of k and ϵ, it was

	Density (kg/m^3)	Boiling point (K)	Freezing point (K)	Formation enthalpy (kJ/mol)
NTO	1380.0	294.3	261.95	-19.564
N_2H_4	1003.7	387.5	275.16	50.434
MMH	870.2	360.8	220.70	54.140
UDMH	786.1	335.5	216.00	48.200

Table 1.3: Propellant properties

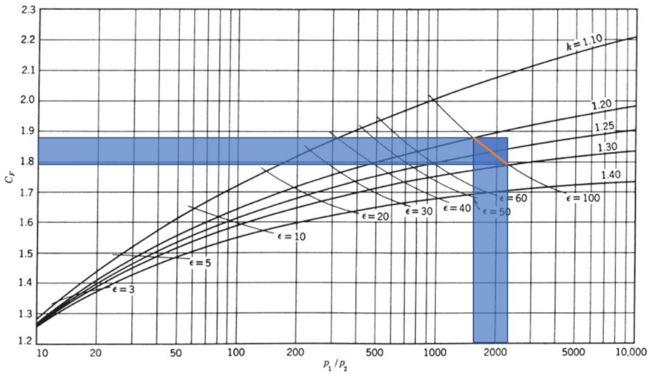


Figure 1.1: Estimated ranges for ϵ and c_f

possible to estimate a range for both (Figure 1.1).

$$c_{f_{vac}} = c_{f_{opt}} + \epsilon \frac{p_c}{p_e} \tag{1.1}$$

$$T_{max} = m_{dry} a_{max} = c_{f_{vac}} p_c A_t \tag{1.2}$$

$$p_c = \frac{a_{max}}{c_{f_{vac}} A_t} m_{dry} \tag{1.3}$$

Starting from those ranges, upper and lower values for $c_{f_{vac}}$ can be calculated (Equation 1.1). The maximum acceleration and a m_{dry} between 1.7kg and 3.99kg imposes a maximum value for thrust (Equation 1.2) and hence a range of possible P_c (Equation 1.3): final results are a range of 50N to 118N for thrust and of 4bar to 10bar for P_c.

1.1 Propellant couple

For the first iteration, the team analysed all three possible propellant couples running the CEA code in rocket problem, assuming infinite area of combustion chamber and freezing point at the combustor. Each couple was studied at various o/f ratios, considering $P_c = 8bar$ and $\epsilon = 100$. From the obtained values of T_c, M and k, the team calculated c^*, c_f and I_{sp} for each case.

CEA Output		**Calculated value**	
T_c (K)	2927	c^* (m/s)	1748.74
M (kg/kmol)	18.564	$c_{f_{opt}}$ (-)	1.8540
c_p (kJ/Kmol)	2.3754	$c_{f_{vac}}$ (-)	1.9105
k (-)	1.2324	I_{sp} (s)	341

Table 1.4: Infinite combustor performance

Since the mission requirement is to produce a given ΔV, the highest I_{sp} will provide the most mass-efficient solution. Simultaneously, the highest propellant average storage density will provide the most volume-efficient solution for the tanks. Based on the combination of these two considerations, the best choice was NTO and N_2H_4 with an o/f of 0.97. The ideal values obtained from this propellant configuration are shown in table 1.4

1.2 Combustion chamber

Combustion chamber geometry plays an important role in rocket efficiency: length and shape affect vaporization efficiency and heat transfer profiles [9]. The reference parameter, L^*, is usually tabulated as depending only on propellant couple, but in reality it is also affected by other factors such as injection, temperature and pressure [10]. Typical values of L^* for NTO - N_2H_4 range between 0.76-0.89m [11].

For this system, where mass and especially volume constraints are tighter than efficiency, the team opted for a combustion chamber length shorter than the ideal value, even though this choice does not grant the maximum combustion efficiency. The team decided to use a cylindrical combustion chamber, a reliable design that integrates very well with the strict volume limitations. Imposing a Mach number of 0.2 for the flow in the chamber, its cross-sectional area would be 3 times the throat area, and the required length would violate the maximum of 9cm (1cm is reserved for injection). However, for throat diameters smaller than 3cm, it is recommended to use a contraction ratio greater than 5 [11]: the team selected this ratio as 9, obtaining an inner diameter of 2.7cm. This gives an L^* of 0.84m, and a Mach number of 0.066 that complies with typical ranges.

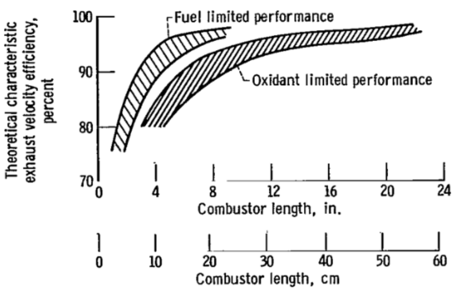


Figure 1.2: c^* efficiency

Even though typical values of L^* are respected, the combustion chamber is still very short, which limits the combustion performance: the estimated c^* efficiency, based on figure 1.2 [7], was 85%. Therefore the actual value of c^* (from the ideal one in table 1.4) is 1486.43m/s .

The combustion chamber structural stress (S) can be calculated as $S = p_c \frac{D}{2} t_w$ using Barlow's formula. Knowing the material's (selected in section 1.4) tensile strength, the minimum wall thickness would be 0.015mm. Therefore, the chamber's thickness was calculated based on thermal considerations, as pressure stresses do not pose a risk.

1.3 Nozzle sizing

To maximize performance given the 10cm-limit for length, the team selected the Rao optimization method, which is still efficient for small rocket nozzles as demonstrated by [13]: the length ratio selected was 60%, the smallest provided by the model. Applying the Rao approach, the team calculated the length of the 60% divergent and the initial and final parabola angles (table 1.5). The

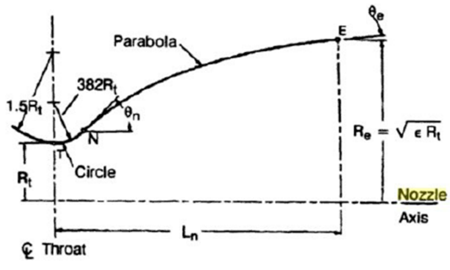
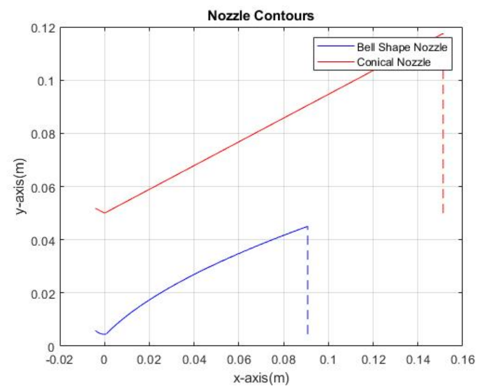


Figure 1.3: Nozzle contour based on Rao method [16]

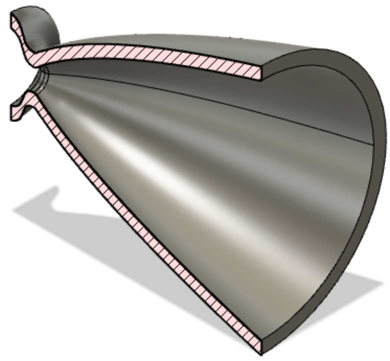
thickness of the nozzle's wall was defined due to thermal considerations (section 1.4).

$R_t(cm)$	0.45	$L_{cone}(cm)$	15.12	$\theta_n(°)$	37
$Re_e(cm)$	4.5	$L_n(cm)$	9.07	$\theta_e(°)$	13

Table 1.5: Nozzle Dimensions



(a) Conical vs Rao nozzle



(b) Cut-away view of nozzle 3D model

Figure 1.4: Nozzle design

Table 1.5 shows the obtained dimensions: these were used to generate 2D (figure 1.4a) and 3D (1.4b) profiles for the nozzle. From the 3D model it was possible to obtain a volume of $89.69cm^3$; and a mass of 402g knowing the material's density (table 1.6).

A rough estimation of the nozzle's 2D losses can be obtained by applying the following formulas [15]:

$$\alpha_{new} = arctan\left(\frac{R_e - R_t}{Ln}\right) \tag{1.4}$$

$$\lambda = 0.5\left(1 + cos\left(\frac{\alpha_{new} + \theta_e}{2}\right)\right) \tag{1.5}$$

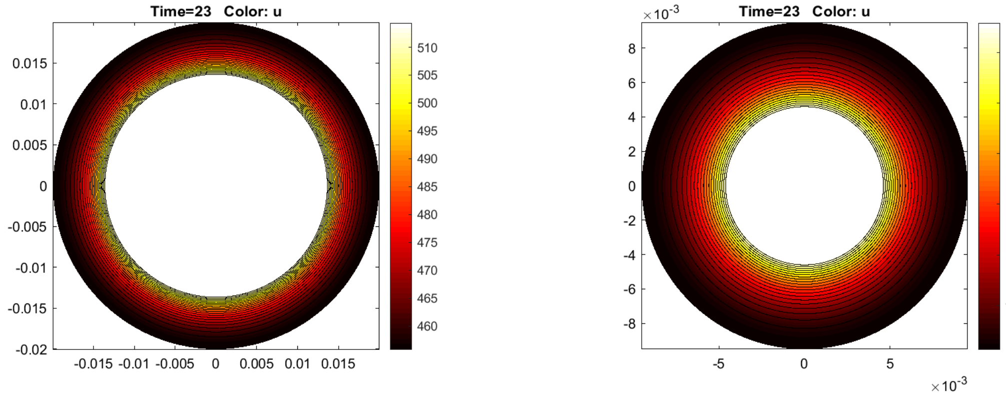
where α_{new} is the new estimation of the nozzle's half angle. A value of $\lambda = 0.9741$ is computed, meaning a ~2.6% reduction in thrust due to 2D losses (accounted for in chapter 3).

1.4 Cooling system

The team selected a high-temperature Titanium alloy, Ti-5Al-2.5Sn-ELI [19], for the combustion chamber and nozzle. This choice is due to a variety of reasons: Titanium alloys are widely used in space applications, and they exhibit some very desirable properties such as a high

Density (kg/m^2)	4480
Ultimate tensile strength (MPa)	724
Thermal conductivity (W/mK)	7.8
Specific heat (J/kgK)	530
β transus temperature (K, nominal)	1311
Melting temperature (K, approximate)	1870

Table 1.6: Properties for Ti-5Al-2.5Sn-ELI



(a) Combustion chamber section

(b) Throat section

Figure 1.5: Simulated temperature distribution at the end of motor firing

strength-to-density ratio; moreover, this particular Titanium alloy has excellent heat capacity, adequate conductivity and exceptional temperature limits.

These properties, combined with the short firing time, allowed the combustion chamber and nozzle to be designed as a heat sink: instead of using a cooling system to manage the heat flux, the material absorbs it for the duration of the firing. To make sure temperatures wouldn't exceed reasonable limits, the team modelled the transient temperature variation in two critical cross-sections: the combustion chamber's (where temperature of the flow is highest) and the throat's (where heat flux is highest).

The convective heat transfer coefficient is given by Barz's model [20]:

$$h = 0.0026\left(\frac{\dot{m}}{A}\right)^{0.8}\left(\frac{c_P}{\mu}\right)^{0.4}\frac{\lambda^{0.6}}{D^{0.2}} \tag{1.6}$$

where A and D refer to the cross-sectional area and cross-sectional diameter respectively and λ is the thermal conductivity of the fluid (obtained from CEA along with most other parameters). Radiative properties of combustion gases are very dependent on fuel and composition, which makes it difficult to estimate the heat exchanged by radiation: the team used an educated guess from [21] for the value of emissivity. Since, as expected [22], radiation does not play an important role in the throat which is the most critical section, this approximation did not compromise the accuracy of the results. The outer surface was modelled as adiabatic in both sections.

The team chose as maximum allowed temperature the material's nominal β transus temperature, as that is the temperature at which the Titanium's structure is permanently altered: this is not necessarily a problem for load-bearing capabilities, however since β transitions generally alter a material's structural properties [23], it was deemed best to err on the safe side.

Thickness of the wall was chosen as 6.5mm at the combustion chamber and 7.5mm at the throat. Figures 1.5a and 1.5b depict the predicted temperature distribution after the longest possible firing: even in the throat, the maximum temperature does not exceed 1100K. This validates the choice of a heat sink design.

Chapter 2

Feeding system and mass configuration

2.1 Injection

Dimension constraints require to limit the size of the injection plate and combustion chamber, which poses a problem since small dimensions lead to less turbulent chambers, hence vaporization and mixing losses [24]. However, an analysis on the effect of injector scale on mixing and combustion [25] shows that detrimental effects such as mixture separation and uneven distribution are absent for very small elements, while they become more relevant as the physical size of the injection element increases.

The team selected like-on-like impinging doublets as the injector type, considering as ruling parameter the use of a storable liquid hypergolic propellant: the configuration has been successfully used [11] and provides the best stability for this type of propellant, opposite to unlike impingement which may initiate high-frequency oscillations with hypergolic propellants [7]. In addition, a pressure drop of at least 1.38 bar across the injector plate is recommended to suppress high-amplitude low-frequency instability [26] (also known as "chugging" [7]) that is common with low mass flow rates and to avoid possible time delays in the atomization, vaporization and combustion processes [27].

	Oxidizer	Fuel
ΔP_{inj} (bar)	1.6	1.6
C_d (-)	0.7	0.7
N (-)	6	6
d_{inj} (mm)	0.493	0.542
$\dot{m}_{inj}$ (kg/s)	0.002809	0.002896
v_{inj} (m/s)	10.659	12.499

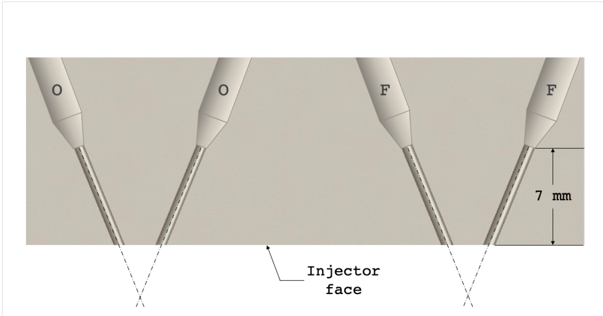
Table 2.1: Injector Plate Parameters

As proposed by [11], an initial estimate of $\Delta P_{inj} = 0.2 P_c$ was selected. The chosen orifice geometry is designed as short tubes with a conical entrance, with 0.5 mm diameter and a corresponding c_d of 0.7 [12]: according to [25], stream separation effects are negligible for this thruster size when injection orifices are small (0.5 mm) and become significant at intermediate diameters (1.5 mm). Additionally, for hypergolic propellants, combustion performance is increased if the resultant momentum vector of each pair of impinging streams is inclined 2° to 5° with respect to the chamber's axis, as it causes re-circulation and better mixing of the liquid propellants along the chamber wall [11]: therefore, asymmetric impingement streams were selected to achieve this effect (figure 2.1).

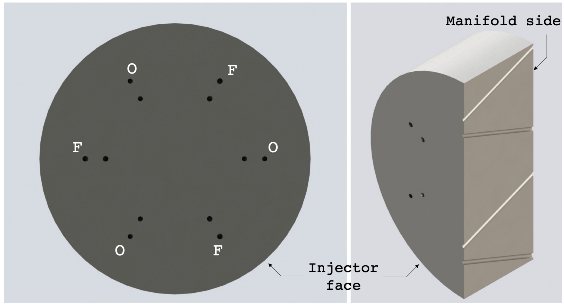
$$\dot{m} = A_{inj} C_d \sqrt{2 \Delta P_{inj} \rho} \tag{2.1}$$

Equation (2.1) was used to size the injector plate, knowing the mass flow rate of the liquid propellant from chapter 1: main characteristics are shown in table 2.1. The same Titanium alloy used for the combustion chamber was selected for the injector structure due to its compatibility with the propellant [28] and the properties of thermal management required to prevent the vaporization of propellants in the injector.

The low propellant mass flow rate requires small cross-sectional area orifices and manifolds, which require high manufacturing precision. Photo-etched thin plates (also referred as "platelets"), which can be bonded together to form a monolithic structure [12], achieve these precision requirements. According to [29], the smallest injector orifice a manufacturer can produce is $2.58 \times 10^{-5}\mathrm{cm}^2$: calculated areas are all above this limit.



(a) Fuel and oxidizer doublets



(b) Face pattern

Figure 2.1: Injection plate

Injector plate design consists of empirical procedures, and the lack of precise correlations between parameters makes it rely on experimental testing to assure a successful design. Water flow tests will be required to determine the injector's actual orifice discharge coefficient, effective pressure drop, distribution patterns, and the quality of atomization. Moreover, hot firing tests are the only method that can fully and reliably evaluate the injector's precise operational characteristics, such as performance, combustion stability, and heat transfer characteristics [11].

2.2 Feed lines

The feeding pipes are regulated by solenoid valves lined with Teflon to prevent deterioration: these components already exist for CubeSats applications and have a mass of about 30g each [30]. The pipes have been designed starting from the mass flow rate data from chapter 3. An initial guess for the tube's diameter was 5mm, from usual values of small liquid propulsion thrusters already available, such as R-6F or R-4D [32]. The team performed preliminary calculations applying equation 2.2 for both oxidizer and fuel separately:

$$\dot{m} = A_{pipe}\rho v_{pipe} \tag{2.2}$$

From this evaluation and knowing the viscosity values of both fluids[33, 5], it was possible to compute the Reynolds number. Even for pipe lengths as low as 5cm, both values of the Reynolds number are high enough to consider the flows turbulent. Therefore, the following formulas [31] have been applied:

	Oxidizer	Fuel
$\dot{m}$ (kg/s)	0.0186	0.0174
v_{pipe} (m/s)	0.686	0.882
μ_{amb} (Pa*s)	4.2×10^{-4}	8.76×10^{-3}
Re_{pipe} (-)	6.91×10^{7}	1.03×10^{6}
λ (-)	0.0035	0.0099
ΔP (Pa)	33.8	38.7

Table 2.2: Feed Pipes flow data

$$\lambda = \frac{0.316}{Re^{0.25}} \tag{2.3}$$

$$\Delta P = \frac{1}{2}\rho v_{pipe}^2 * (\lambda \frac{L_{pipe}}{d_{pipe}}) \tag{2.4}$$

Table 2.2 summarises the results: as it was expected due to the shortness of the tubes and the very low mass flow rate, distributed pressure losses are small.

The material chosen for the pipes is aluminium [28],thanks to its low density and mechanical properties that can meet the requirements of the mission. The team estimated a thickness of 1mm for the pipes and about 40cm of total pipe length required for all the lines (section 2.4 contains a scheme of the tanks' distribution). After having calculated the total volume of the pipes, their mass is estimated to be about 37.3g. Therefore the total mass of the feeding system is ~160g, taking into account the pipes and four valves; two located between the N_2 tank and each propellant and two between those and the injector plate.

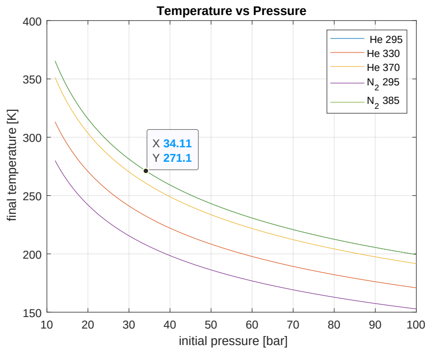
2.3 Propellant storage

Given the small size of the system, a blowdown pressure-fed system was designed for propellant storage and feeding, using a single N_2 tank to pressurise both propellant tanks. The team selected spherical tanks, with each of the two propellant tanks being oversized by 2% to account for ullage volume, and chose graphite ($\sigma_y = 895MPa$, $\rho =$ 1550kg/m^3) as tank material since it is lightweight and highly stress-resistant. It was also assumed to apply a coating or liner to the inside of the tank to protect it from corrosion, since the selected propellants are highly aggressive.

The propellant mass needed was calculated knowing the mass ratio (discussed in chapter 3) and assuming a dry mass. Together with the o/f ratio and each propellant density, the required tank volumes were obtained. Including safety factors and losses, the team computed the radius of the tanks and their thickness applying Humble's formula with a safety factor of 2.5 (corresponding to a burst pressure of 25bar):

$$t = \frac{P_i f_s r}{2\sigma_y} \tag{2.5}$$

Given the small amount of propellant and the narrow temperature range for their liquid phase, the primary consideration for selection of the pressurising gas was its final temperature once expanded, which could not be below 265K: Helium proved inadequate, which is why Nitrogen was selected. Even then, the final design evaluation resulted in selecting an initial temperature for the pressurising gas of 385K at a pressure of 34bar, meaning that there is the need for dedicated thermal control for the heating of the pressurisation gas. To find the volume of pressure gas, an iterative approach had been used with 5% allowed error.



After obtaining the tank and pressure systems masses, the dry mass was recalculated to repeat the process with a more accurate value. Final results are shown in table 2.3. Final masses are discussed in section 2.4.

	Oxidizer	Fuel	Pressurisation gas
Tank mass (dry, g)	2.6	3.4	14.1
Tank mass(wet, g)	383.2	395.7	28.7
Volume (cm^3)	311.1	400.9	500.7
Radius (cm)	4.203	4.574	4.926
Thickness (mm)	0.0762	0.0829	0.3022
Pressure (bar)	10	10	34

Table 2.3: Final values for the storage system

2.4 Center of mass

Calculations for the center of mass (CM) of the satellite are essential to ensure no torques are being applied on the spacecraft during firing. To compute it, it is necessary to tally up all the masses of the system: the empty mass for each cube structure is 145g [34], while masses for all components were estimated in the previous sections and are reported in table 2.4.

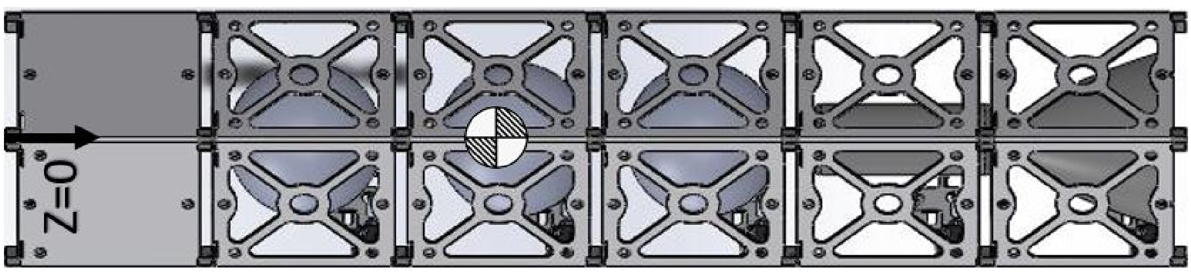


Figure 2.2: Schematic configuration and CM location. From left to right: Payload, pressurising gas, oxidizer, fuel, combustion chamber and nozzle

Defining a reference system with z-axis along the axis of the spacecraft and x- and y- axes orthogonal to it (see figure 2.2), the team placed all components along the

	Initial values			Final values		
	CM (cm)	Mass (kg)	kg*cm	CM (cm)	Mass (kg)	kg*cm
Payload	5	1.33	6.65	5	1.33	6.65
Cube 1 empty	15	0.145	2.175	15	0.145	2.175
N2 tank	15	0.0141	0.2115	15	0.0141	0.2115
N2 mass	15	0.0146	0.219	25	0.0146	0.365
Cube 2 empty	25	0.145	3.625	25	0.145	3.625
Valves and pipes	25	0.160	4.00	25	0.160	4.00
Oxidizer tank	25	0.0026	0.065	25	0.0026	0.065
Oxidizer	25	0.3806	9.515	-	0	-
Cube 3 empty	35	0.145	5.075	35	0.145	5.075
Fuel tank	35	0.0034	0.119	35	0.0034	0.119
Fuel	35	0.3923	13.731	-	0	-
Cube 4 empty	45	0.145	6.525	45	0.145	6.525
Injection plate	40.5	0.05352	2.1676	40.5	0.05352	2.1676
Combustion chamber	45.5	0.2758	12.5489	45.5	0.2758	12.5489
Cube 5 empty	55	0.145	7.975	55	0.145	7.975
Nozzle	56.2	0.4018	22.581	58	0.4018	23.3044
Total	25.89	3.75	97.18	25.10	2.98	74.81

Table 2.4: Initial and final center of mass calculation

axis of symmetry, to ensure that the CM would be on the z-axis at all times. Non-central masses, like the feed lines, were considered small enough to be lumped at the centre of the satellite. The initial and final positions of the CM are $z_i = 25.89cm$ and $z_f = 25.10cm$ respectively. This variation is very small, and indeed it can be considered negligible given the overall length of the satellite.

Chapter 3

Final performance evaluation

Once $P_c = 8bar$ and $\epsilon = 100$ were defined, the team ran the CEA code in rocket problem assuming a finite combustor area with a contraction ratio of 9. The same calculations for c^*, c_f and I_{sp} were performed, obtaining similar results (table 3.1)

CEA Output		Calculated value	
T_c (K)	2911	C^* (m/s)	1743.10
M (kg/kmol)	18.563	$Cf_{optimal}$ (-)	1.8541
C_p (kJ/Kmol)	2.3736	Cf_{vacuum}	1.9105
k	1.2323	I_{sp}	339

Table 3.1: Finite combustor performance

To calculate a more realistic engine performance, we estimated efficiencies to account for losses in the combustion chamber and the nozzle: from the computed c^* efficiency of 85%, the team calculated the new I_{sp} of 289s, and from that a mass ratio of $n_m = 1.2361$ applying Tsiolkovsky's equation.

Real nozzles experience performance losses due to lower velocities at the boundary layer [2], and this effect is more significant in smaller nozzles due to smaller throat areas [17, 18]. Using the correlation method introduced by Spisz [17], the team calculated that viscous effects and hot gas effects (enthalpy losses due to heat transfer with the wall and dissociation) combined account for a loss in c_f of ~4%. It should be noted, however, that the references presented different values of A_t and $\dot{m}$ for which losses are likely to be greater than in this case: 4% may be an overly conservative value, but the team judged it ideal as a blanket term to account for other losses for which no analytical derivation was present in the literature.

To this, another factor of 2% was added to account for losses in the convergent part of the nozzle, by interpolating values that were tabulated in [2] for similar conditions. Finally, adding 2D losses already calculated in section 1.3 gives an overall thrust efficiency decrease of 9%, translating to an actual thrust of 87.51N. This amount of efficiency loss is indeed similar to those given by [2].

Table 3.2 summarises the final parameters of merit for the propulsion unit: while the need for miniaturization severely impacted efficiency, the end result is still a very effective propulsion unit given its size. It achieves the design requirements and complies with constraints. Moreover, the use of hypergolic propellants not only simplifies the system, it also allows multiple ignitions of the engine, although thrust is not throttleable nor vectorable. The decision to keep every component inside 1U means additional modules (for example, additional fuel tanks) can be added to the assembly with little redesign required.

I_{sp} (s)	289
Mass ratio, n_m (-)	1.2361
Thrust (N)	87.51
$\dot{m}$ (kg/s)	0.03092
Burning time (s)	23
Total impulse (N.s)	1996

Table 3.2: Final performance

3.1 Off-design evaluations

To conclude this study, the team would like to highlight some possible failure modes the system could be subjected to, and suggest possible mitigation strategies or testing requirements that could help in avoiding them.

Structural failures due to excessive pressure, such as bursting of the tanks or over-pressure in the combustion chamber, are not expected since system pressure is fairly low and both the tanks and the combustion chamber were designed with high safety factors against that risk. Due to the slenderness of the design and the pure axiality of the loads, however, buckling effects must to be evaluated as a possible failure mode through a dedicated structural analysis, as there is no way to know the impact they would have on a mission.

In case of leakage of the propellants due to valve malfunctioning, they may freeze in the injector manifolds, blocking the flow [36]. This may cause difficulties in ignition, especially for hypergolic couples. If later data found this to be a major concern, prevention could be implemented by designing a heating system for the valves and feed lines. A heating system would also need to be designed for the tanks if the specific orbit of the satellite resulted in an equilibrium temperature lower than the predicted one [37], as propellant freezing is a major concern with this design: loss of thrust may be complete in the worst cases.

As the injector plate would be one of the most complex parts of the design, it is expected that several failure modes would be associated with it [35]: incomplete vaporization can occur due to errors in the injector pattern design, and the short chamber length could lead to o/f ratio variation and the possible ejection of unburnt propellant. Moreover, inadequate heat dissipation at the injector head could lead to vaporization of the propellants inside the injector orifices [24]. Ignition and shut-off transients will need to be carefully studied, as bad designs can lead to instabilities due to a severe or insufficient pressure drop across the injector plate [11], which may lead to uneven, oscillating thrust and pressure buildup, although engine explosion is unlikely due to the very high margins of safety used for the chamber.

To conclude, we believe this design to be efficient, versatile and reliable. The study proved miniaturized bipropellant propulsion unit to be a viable solution for CubeSats, and it is our belief that it could be built with current technology.

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Thrust Optimized Parabolic Contours (Rao's approach)

A Rao parabolic nozzle can be fully defined geometrically by three curves, its length and throat radius [14]. The first two circular curves describe the geometry of the contour before and after the nozzle throat (as it can be observed in figure 1.3). The final curve is essentially the equation of a parabola which defines the divergent section of the nozzle. The Rao coefficients used to specify the curvature of the first two sections are 1.5Rt and 0.382Rt respectively, as it can be seen in figure 1.3 [16]. The equations utilized are summarized below [14]:

$$L = \frac{K\left(\sqrt{\epsilon}-1\right)R_t}{tan(\theta_e)} \tag{1}$$

$$x^2 + (y-(R_t+1.5R_t))^2 = (1.5R_t)^2 \tag{2}$$

$$x^2 + (y-(R_t+0.382R_t))^2 = (0.382R_t)^2 \tag{3}$$

$$x = ay^2 + by + c \tag{4}$$

where 1 represents the length of the nozzle's divergent section, 2 and 3 are the equations of the two curves and 4 is the general equation of a parabolic profile. Solving 2 and 3 for y, one obtains:

$$y_1 = -\sqrt{(1.5R_t)^2 - x^2} + 2.5R_t \tag{5}$$

$$y_2 = -\sqrt{(0.382R_t)^2 - x^2} + 1.382R_t \tag{6}$$

It is important to mention that equations 5 and 6 give two solutions each: the solutions to be selected to get a concave nozzle profile are the ones whose second derivative is positive. Moreover, equations 5 and 6 can be used to determine the radius of the nozzle for any value of x.

To define the x coordinate of the first curve, which is the one defining the profile between the combustion chamber and the nozzle throat, the following equation is applied:

$$x_1 = -1.5R_t sin(\theta_1) \tag{7}$$

In addition, the x coordinate at point N along the nozzle must be specified (see figure 1.3). Specifically, the first order derivatives at that point, where the second curve at the right-hand side of the nozzle throat coincides with the starting point of the parabola, are obtained making possible the computation of the parabola's coefficients shown in equation 4.Therefore, x_N is computed as shown below:

$$x_N = 0.382 R_t sin(\theta_N) \tag{8}$$

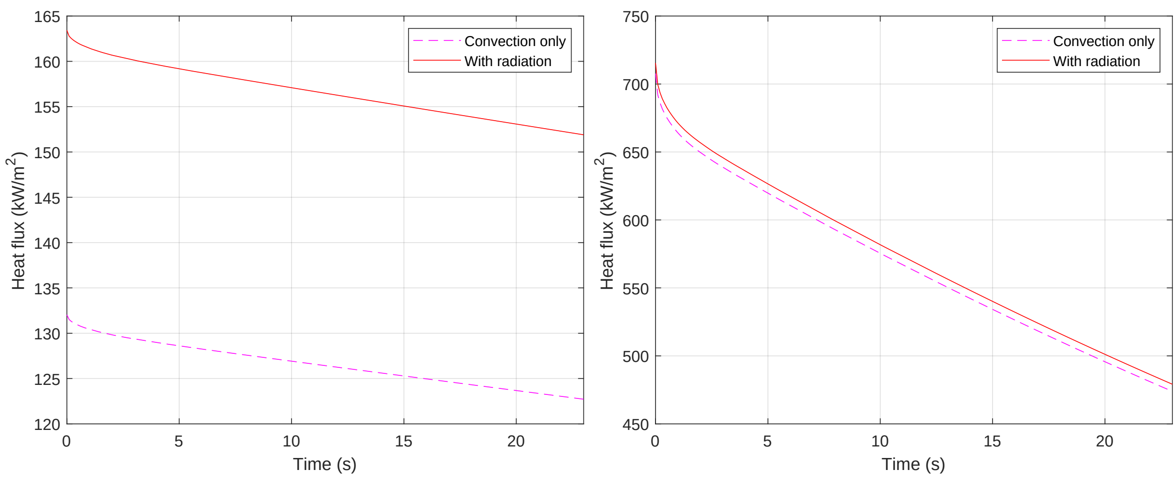
Finally, this last system of equations gives the coefficients of the parabola:

$$\begin{bmatrix} 2R_N & 1 & 0 \\ 2R_e & 1 & 0 \\ R_N^2 & R_N & 1 \end{bmatrix} \begin{Bmatrix} a \\ b \\ c \end{Bmatrix} = \begin{Bmatrix} \frac{1}{tan(\theta_N} \\ \frac{1}{\theta_e} \\ x_N \end{Bmatrix} \tag{9}$$

The equations were implemented in Matlab, generating the 2D profiles of both the conical and its corresponding bell-shape nozzle figure (1.4a). The 2D coordinates defining the shape of the latter were then imported into a CAD software, namely "Fusion 360", to obtain a 3D visualization of the given configuration (figure 1.4b).

Modelling of radiative heat exchanges

Data about transient radiation exchanges in rocket motors is very scarce, which is why for the purpose of this analysis some simplifying assumptions were used: first of all, seeing that Hydrazine-NTO combustion produces fairly clean exhaust, without any major presence of soot, the team chose a fairly low value for emissivity of $\epsilon = 0.2$. This value was chosen as an educated guess, however it is worth noting that other values were tested and the change did not have a major effect on the behaviour of the system. The emissivity for the metal surface, both internal and external, of the combustor and nozzle was chosen as $\epsilon = 0.5$, also an educated guess, based on typical values for polished metals.



(a) Combustion chamber (b) Throat

Figure 1: Heat flux during motor firing

Modelling the exchange in its most accurate form would require the use of a dedicated thermal CFD software to evaluate the emission from a participating media (the exhaust) flowing in the combustor and nozzle. For the purpose of this analysis, we deemed this unnecessary and time-consuming: instead, to calculate the radiative exchanges the team modelled the gas's emission as the emission of a solid column of radius $r = \frac{r_{duct}}{2}$ in the center of the duct, whose emissivity was the emissivity of the gas. This approach leaves much to be desired in terms of accuracy, however it is sufficiently accurate for this analysis.

The data indeed confirmed the expected trend: radiative heat exchange plays a significant role in the combustion chamber, where temperatures are higher and convection less prominent (figures 1a and 2a), but in the throat, where convective heat exchange is at a maximum, it is almost completely negligible (figures 1b and 2b).

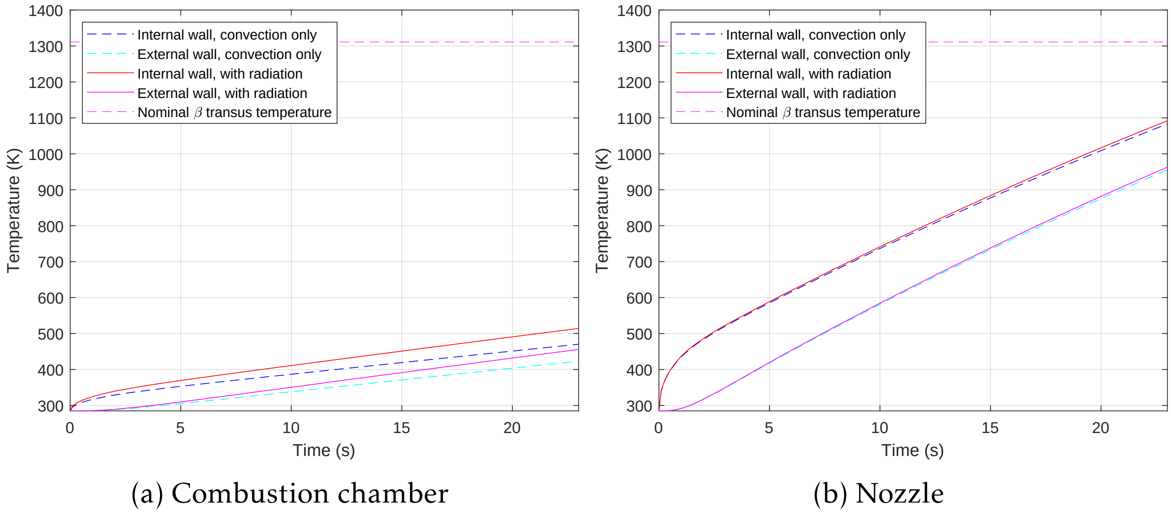


Figure 2: Temperatures of the inner (solid line) and outer (dashed line) surfaces during motor firing